

It's easy to see that $p_1q_2 + p_2q_1$ is an integer and q_1q_2 is an integer, $q_1 \neq 0$ and $q_2 \neq 0$. $\therefore q_1q_2 \neq 0$. By Definition 2, r/s is a rational number. $\square$

Example 9: Prove that if n is an integer and n^2 is odd, then n is odd.

Soln: We will prove this by contraposition.

Let n be an arbitrary even integer. By Defn 1, there exists an integer K such that $n = 2K$. $\therefore n^2 = (2K)^2 = 4K^2 = 2(2K^2)$. $2K^2$ is an integer.

By Defn 1, n^2 is an ^{even} integer. $\square$

1.7.7 Proofs by Contradiction

• Suppose we want to prove that a statement p is true. Furthermore, suppose that we can find a contradiction q such that $\neg p \rightarrow q$ is true. Because q is false, but $\neg p \rightarrow q$ is true, we can conclude that $\neg p$ is false, which means that p is true. How can we find a contradiction q that might help us ^{prove} that p is true in this way?

• Because the statement $p \wedge \neg p$ is a contradiction whenever p is a proposition, we can prove that p is true if we can show that $\neg p \rightarrow (p \wedge \neg p)$ is true for some proposition p . Proofs of this type are called proofs by contradiction.

Example 10: Show that at least four of any 22 days must fall on the same day of the week.

Soln: Suppose, for the sake of contradiction, at most 3 of any 22 days falls on the same day of the week. Since there are 7 days in a week, this implies we can have at most 21 days chosen, which is a contradiction to the fact that we are given 22 days. $\therefore$ At least four of any 22 days must fall on the same day of the week. $\square$